Ian Lingo

ian.lingo21@gmail.com | ianlingo.com | linkedin.com/in/ianlingo/ | github.com/ian8912

Education

Texas State University San Marcos, TX
B.S. in Computer Science *Graduated: May 2026*

- **GPA**: 3.58
- **Minors**: Applied Mathematics, Mass Communication
- **Honors**: Cum Laude, IEEE-HKN (Eta Kappa Nu), Texas State Honors College
- **Coursework**: Computer Architecture, Digital Logic, Operating Systems, Computer Vision, Data Structures & Algorithms, Systems Security, Software Engineering, Mechanics

San Jose State University - Open University San Jose, CA
Advanced Python Programming *Jun. 2026 – Aug. 2026*

Technical Skills

Programming Languages: C++, Python, Java, JavaScript, TypeScript, SQL
Embedded Hardware/Interfaces: Jetson Orin Nano, Teensy 4.1, Arduino, UART, USB Serial, PWM, I2C, GPIO, Encoders, ToF Sensors, FPGA Fundamentals
Robotics/Computer Vision: ROS2, ZED 2i, RPLIDAR A1, OpenCV, YOLOv11, AprilTag
Systems/Development: Ubuntu Linux, Git, GitHub Actions, CI/CD, Docker, Gradle, AWS, OCI
Testing/Software: JUnit 5, Mockito, Testcontainers, Spring Boot, REST APIs, PostgreSQL
Additional Tools: React, Next.js, Claude Code, GitHub Copilot
Certificates: Google Data Analytics, Oracle Cloud (OCI) Foundations, AWS Certified Developer - Associate (In Progress)

Projects

MARS Robot | *ROS2, Embedded Systems, Python, C++, YOLO, ML, Computer Vision* Sep. 2025 – May 2026

- Developed **C++ PWM motor-control firmware** on a **Teensy 4.1** microcontroller and interfaced it with **Python ROS2 software** running on an NVIDIA Jetson Orin Nano via USB serial communication.
- Integrated and debugged **embedded motor control, sensor inputs, and teleoperation** across Linux-based ROS2 nodes and hardware interfaces, supporting mecanum mobility and Xbox controller operation.
- Engineered a **YOLOv11 Nano computer-vision pipeline** using a **229-image annotated dataset** and ZED 2i camera for real-time industrial object detection; contributed to human-tracking functionality.
- Directed a **17-member** IEEE RAS engineering team in Agile Scrum to develop an indoor service robot platform using NVIDIA Jetson Orin Nano, ZED 2i, RPLidar A1, ToF sensors, CAD, and embedded motor control.

Pico - IEEE Region 5 Robotics Competition | *C++, Python, PWM, I2C, Encoders, AprilTag* Dec. 2025 – Mar. 2026

- Developed **C++ mecanum movement-control software** using PWM motor control, encoder feedback, and I2C ToF sensing; implemented **Python AprilTag-based perception** to support robotic navigation and competition tasks.
- Collaborated on a **5-member engineering team** to iteratively debug and improve robot performance, achieving **2nd place out of 25 universities** at the IEEE Region 5 Robotics Competition.

Chess Game Application | *Java, Spring Boot, PostgreSQL, Docker, AWS EC2, JUnit 5* Jul. 2024 – Present

- Engineered a **full-stack** Java chess app (Swing client + Spring Boot REST API) with a multi-module Gradle build and a from-scratch **AI opponent** using negamax, alpha-beta pruning, and quiescence search.
- Built an Elo-rated leaderboard with a Spring Boot REST API backed by PostgreSQL; containerized with Docker, **deployed to AWS EC2**, covered by **20 JUnit 5 tests** using Mockito and Testcontainers in a **GitHub Actions CI** pipeline.

Experience

Geek Squad - Advanced Repair Technician Jun. 2026 – Present
Best Buy *San Jose, CA*

- Diagnose and troubleshoot **hardware/software failures** across Windows, macOS, and Linux; perform **post-repair functional testing**, document Apple GSX workflows, and resolve device reliability issues as an Apple-certified technician.

Director, Vice President, Treasurer, and Co-Founder Aug. 2024 – May 2026
Boko Hackathon, IEEE, IEEE-HKN, and IEEE RAS Chapter *San Marcos, TX*

- Co-founded IEEE Robotics & Automation Society (RAS), growing the chapter to **100+ members** and supporting student engineering initiatives.
- Co-directed an AI and cybersecurity hackathon with **100+ participants**, served as a technical code judge, and raised **$12,000+** from Google, Dell, BrainTrust, UFCU, IEEE, and Charles Schwab via sponsorship outreach.

Team Lead and Learning Ambassador (Operations Associate) Jan. 2021 – Jul. 2025
Amazon *San Antonio, TX and Austin, TX*

- Led and trained a **10+ member** outbound operations team, monitoring supply-chain throughput of **10,000+ packages daily** and implementing process improvements that reduced truck dispatch time by **50%**.